413/613 Fall 2026 - Applied Statistics & Data Science I - Lecture 16

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Recall

In Lectures 14–15, we introduced regularization and model-selection criteria for controlling model complexity. For linear regression, ridge regression solves

$$\widehat{\beta}_\lambda^{\text{ridge}} = \arg\min_\beta \left\{ \|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2 \right\},$$

with the tuning parameter λ chosen using a model-selection procedure such as cross-validation. We now study how the choice of penalty changes the fitted model and why an L_1 penalty can perform *feature selection*.

1 Regularization as a Family of Models

Consider the regression model

$$y = \beta_0 \mathbf{1} + X\beta + \varepsilon, \qquad X \in \mathbb{R}^{n \times p}, \qquad \beta \in \mathbb{R}^p.$$

A general regularized estimator has the form

$$(\widehat{\beta}_0, \widehat{\beta}_\lambda) = \arg\min_{\beta_0, \beta} \left\{ \|y - \beta_0 \mathbf{1} - X\beta\|_2^2 + \lambda g(\beta) \right\}, \qquad \lambda \geq 0,$$

where the intercept is not penalized. Two important choices are

$$\boxed{g(\beta) = \|\beta\|_2^2 = \sum_{j=1}^p \beta_j^2} \qquad \text{(ridge)}$$

and

$$\boxed{g(\beta) = \|\beta\|_1 = \sum_{j=1}^p |\beta_j|} \qquad \text{(Lasso)}.$$

The parameter λ controls the tradeoff between fitting the observed data and constraining the coefficient vector:

$$\lambda = 0 \implies \text{OLS}, \qquad \lambda \uparrow \implies \text{stronger shrinkage}, \qquad \lambda \to \infty \implies \widehat{\beta}_\lambda \to 0.$$

Thus, varying λ generates a family of candidate models. Model selection can therefore be viewed as choosing an appropriate value of λ. The main distinction is qualitative. Ridge makes coefficients small, while Lasso can make coefficients exactly zero. Define the number of active coefficients by

$$\|\widehat{\beta}\|_0 := \#\{j : \widehat{\beta}_j \neq 0\}.$$

A model is called *sparse* when $\|\widehat{\beta}\|_0 \ll p$.

2 A Bayesian Interpretation: Why These Penalties?

Assume Gaussian regression errors,

$$Y \mid X, \beta_0, \beta \sim \mathcal{N}(\beta_0 \mathbf{1} + X\beta, \sigma^2 I_n).$$

Then

$$-\log p(y \mid X, \beta_0, \beta) \equiv \frac{1}{2\sigma^2} \|y - \beta_0 \mathbf{1} - X\beta\|_2^2,$$

where $\equiv$ denotes equality up to an additive constant independent of the parameters. Maximum a posteriori estimation uses

$$p(\beta_0, \beta \mid y, X) \propto p(y \mid X, \beta_0, \beta)\, p(\beta),$$

so

$$(\widehat{\beta}_0, \widehat{\beta})_{\text{MAP}} = \arg\min_{\beta_0, \beta} \left\{ -\log p(y \mid X, \beta_0, \beta) - \log p(\beta) \right\}.$$

We take the intercept prior to be flat and place independent priors on the slope coefficients.

2.1 Gaussian prior $\Rightarrow$ ridge

Suppose

$$\beta_j \overset{\text{iid}}{\sim} \mathcal{N}(0, \tau^2).$$

Then

$$-\log p(\beta) \equiv \frac{1}{2\tau^2} \|\beta\|_2^2.$$

Multiplying the MAP objective by $2\sigma^2$ gives

$$\boxed{(\widehat{\beta}_0, \widehat{\beta})_{\text{MAP}} = \arg\min_{\beta_0, \beta} \left\{ \|y - \beta_0 \mathbf{1} - X\beta\|_2^2 + \lambda \|\beta\|_2^2 \right\}, \qquad \lambda = \frac{\sigma^2}{\tau^2}.}$$

Thus ridge corresponds to the prior belief that coefficients are concentrated near zero.

2.2 Laplace prior $\Rightarrow$ Lasso

Suppose instead

$$p(\beta_j) = \frac{1}{2b} \exp\left(-\frac{|\beta_j|}{b} \right).$$

Then

$$-\log p(\beta) \equiv \frac{1}{b} \|\beta\|_1,$$

and therefore

$$\boxed{(\widehat{\beta}_0, \widehat{\beta})_{\text{MAP}} = \arg\min_{\beta_0, \beta} \left\{ \|y - \beta_0 \mathbf{1} - X\beta\|_2^2 + \lambda \|\beta\|_1 \right\}, \qquad \lambda = \frac{2\sigma^2}{b}.}$$

The Laplace density is sharply peaked at zero and has heavier tails than a Gaussian density with comparable variance.

3 Why Does Lasso Produce Sparse Solutions?

For every suitable value of the penalty parameter λ, the penalized problem can be written in an equivalent constrained form. Ridge corresponds to

$$\min_{\beta_0, \beta} \|y - \beta_0 \mathbf{1} - X\beta\|_2^2 \qquad \text{subject to} \qquad \|\beta\|_2 \le t,$$

while Lasso corresponds to

$$\min_{\beta_0, \beta} \|y - \beta_0 \mathbf{1} - X\beta\|_2^2 \qquad \text{subject to} \qquad \|\beta\|_1 \le t.$$

The value of t decreases as the penalty becomes stronger. In two dimensions, the L_2 constraint is a circle, while the L_1 constraint is a diamond. The least-squares contours expand around the unconstrained OLS

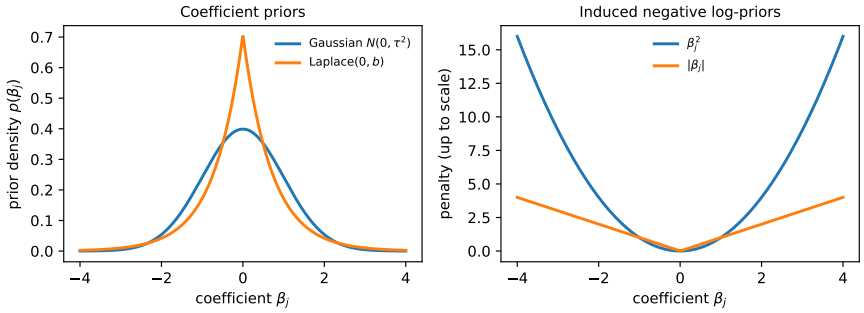


Figure 1: Gaussian and Laplace coefficient priors and the corresponding negative log-priors. A Gaussian prior induces a quadratic L_2 penalty; a Laplace prior induces an absolute-value L_1 penalty.

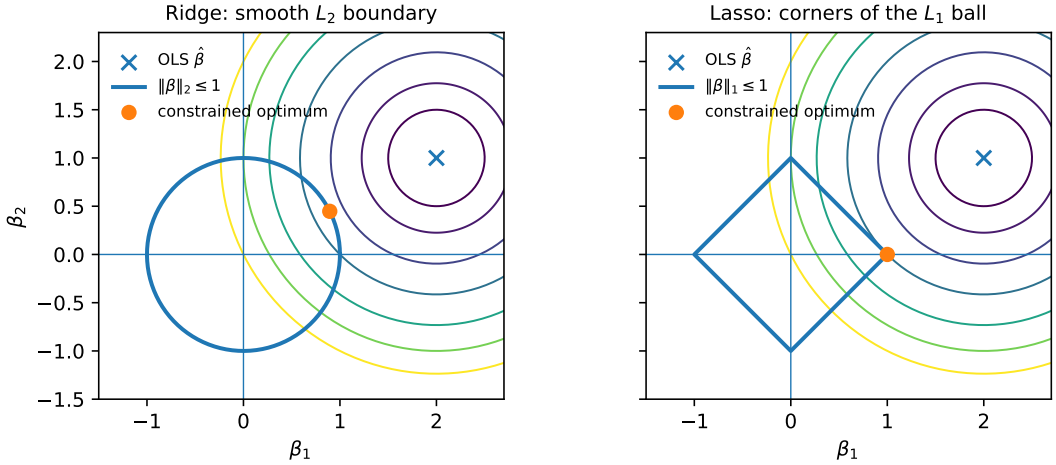


Figure 2: Geometric interpretation of ridge and Lasso. The smooth L_2 boundary typically produces a solution with both coordinates nonzero. The L_1 ball has corners on the coordinate axes, so the first point of contact can occur at a corner, producing an exactly zero coefficient.

solution until they first touch the feasible region.

The geometry explains the difference between shrinkage and sparsity:

ridge: $\widehat{\beta}_j$ is typically small but nonzero versus Lasso: some $\widehat{\beta}_j = 0$.

Hence Lasso performs coefficient estimation and feature selection simultaneously. More generally, one can consider penalties based on L_q quantities,

$$\sum_{j=1}^{p} |\beta_j|^q.$$

The cases $q = 2$ and $q = 1$ give ridge and Lasso, respectively. The special role of $q = 1$ is that the feasible region is convex but has axis-aligned corners; these corners are responsible for sparse solutions.

3.1 Correlated predictors

Sparsity does not automatically imply a uniquely meaningful selected feature set. If two predictors are strongly correlated, they can explain nearly the same variation in the response. Ridge tends to distribute the fitted effect across correlated predictors, whereas Lasso may retain one and set another to zero. Which variable is selected can be unstable under small perturbations of the data. Therefore, selected variables should be interpreted together with the correlation structure of X.

4 Feature Scaling Before Regularization

Regularization acts directly on coefficient magnitudes. Consequently, the penalty depends on the units in which the predictors are measured. If one predictor is measured on a scale of 10^4 and another on a scale of 1, their fitted coefficients can have very different magnitudes even when the predictors are comparably important. For each nonconstant predictor j, define

$$\bar{x}_j = \frac{1}{n}\sum_{i=1}^{n} x_{ij}, \qquad s_j = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2},$$

and standardize

$$\boxed{\widetilde{x}_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j}.}$$

Then each standardized predictor has sample mean zero and sample standard deviation one under the convention above. Ridge and Lasso should generally be fit after this transformation; otherwise the penalty can favor or suppress variables simply because of their units.

5 Case Study: Predicting Episode Ratings

The in-class example treated the rating of a television episode as a response and constructed a high-dimensional feature vector from quantities such as

dialogue counts, character indicators, writers/directors, season indicators.

For episode i, let y_i denote its rating and let $x_i \in \mathbb{R}^p$ contain the engineered features. The model is

$$y_i = \beta_0 + x_i^T \beta + \varepsilon_i.$$

The practical workflow is:

1. construct the design matrix X from the candidate features;
2. standardize the columns of X;
3. fit OLS, ridge, and Lasso models;
4. choose λ for ridge/Lasso using cross-validation or another model-selection criterion;
5. compare predictive performance and inspect the fitted coefficients.

In the class example, the full model contained 87 candidate coefficients. OLS and ridge retained essentially all of them, whereas the selected Lasso model retained only 10 nonzero coefficients. This illustrates the practical value of sparsity: a model can remain predictive while identifying a substantially smaller set of features.

However, a nonzero Lasso coefficient indicates a predictive association conditional on the fitted model; it does *not* establish a causal effect. This distinction becomes especially important when features encode demographic or socially sensitive information.

6 Choosing the Regularization Parameter

The value λ is a hyperparameter. For a grid

$$\lambda_1, \ldots, \lambda_M,$$

one may fit the corresponding models and choose λ using cross-validation. The training error alone is not sufficient: OLS ($\lambda = 0$) always achieves training RSS no larger than the penalized fits. The role of validation is to determine whether the reduction in model complexity improves prediction on unseen data.

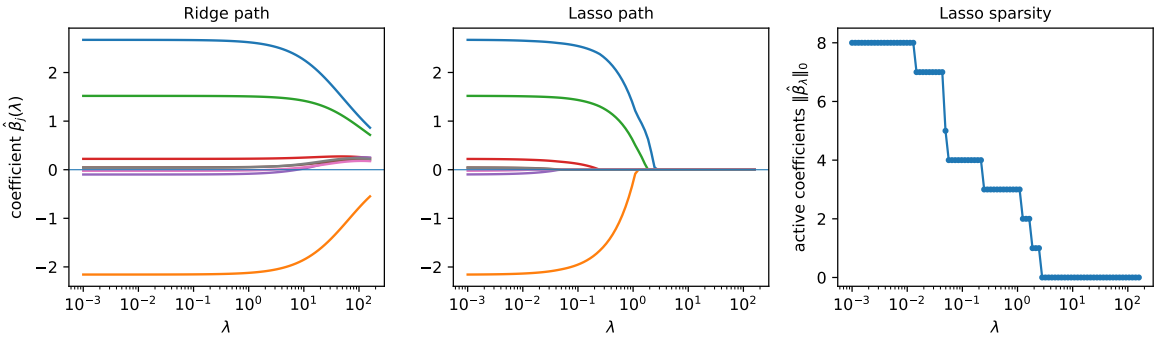


Figure 3: Regularization paths on standardized synthetic predictors. Ridge shrinks all coefficients smoothly. Lasso drives many coefficients exactly to zero; the right panel shows the corresponding number of active coefficients.

7 Regularization Paths

A regularization path records how each fitted coefficient changes as the penalty parameter varies:

$$\lambda \longmapsto \widehat{\beta}_j(\lambda), \qquad j = 1, \ldots, p.$$

For centered predictors and response, ridge has the closed-form path

$$\widehat{\beta}_\lambda^{\text{ridge}} = (X^T X + \lambda I_p)^{-1} X^T y.$$

As λ increases, the coefficients shrink continuously toward zero. For finite λ, ridge coefficients are generally nonzero. Lasso has no analogous closed-form expression for a general design matrix. Its path nevertheless has a distinctive property: coefficients can become exactly zero. Define the active set and its size by

$$A(\lambda) = \{j : \widehat{\beta}_j(\lambda) \neq 0\}, \qquad s(\lambda) = |A(\lambda)| = \|\widehat{\beta}(\lambda)\|_0.$$

Stronger Lasso regularization generally produces a smaller active set, although with correlated predictors individual variables can enter or leave the active set along the path.

Regularization paths provide a useful diagnostic even before a final value of λ is chosen. They show which coefficients are stable over a wide range of penalty strengths and which features disappear quickly as the model is made sparser.

8 Summary

- Ridge and Lasso solve the same data-fitting problem but impose different structural assumptions on β.
- A Gaussian coefficient prior leads to ridge MAP estimation; a Laplace prior leads to Lasso MAP estimation.
- The corners of the L_1 constraint explain why Lasso can set coefficients exactly to zero.
- Predictor standardization is essential before comparing or penalizing coefficient magnitudes.
- Lasso provides a sparse predictive model, but selected variables must be interpreted carefully in the presence of correlated or sensitive features.